四个代码重写专家模型实验报告

Qwen3-4B LoRA

Formal v1 |

1 研究目的与结论

Qwen3-4B

LoRA Formatting Identifier Local Structure Control Flow 50.34 27.06 15.37 33.90 98.88%

2 四个专家的职责分工

Qwen3-4B LoRA

Formatting			AST
Identifier			
Local Structure			
Control Flow		if	

Formatting Identifier Local Structure Control Flow

3 数据与训练设置

problem_id train validation test Qwen3-4B 4-bit QLoRA LoRA rank 16 LoRA alpha 32 2×10^{-4} assistant-only causal language modeling loss

		Train	Validation	Test	checkpoint
Formatting		5,892	736	737	checkpoint-900
Identifier		5,835	722	728	checkpoint-700
Local Structure		11,085	1,385	1,386	checkpoint-600
Control Flow		4,393	567	534	checkpoint-500

3.1 专家训练数据集记录

“ ” problem_id problem_id Base/LoRA

Formatting	BigCode CommitPackFT Python	→ AST	STYLER Learning Natural Coding Conventions
Identifier	CodeSearchNet Python	DOBF / → -	DOBF
Local Structure	CodeSearchNet Python	Extract/Inline /	NatGen

Control Flow	DeepMind CodeContests Python	guard clause conditional rewrite loop early return break/continue	NatGen ContraCode

Formatting bigcode/commitpackft Identifier Local Structure code-search-net/code_search_net Control Flow deepmind/code_contests

Formatting	teacher_training/data/processed/formatting/train.jsonl	teacher_training/data/processed/formatting/validation.jsonl	5,892	736
Identifier	teacher_training/data/processed/identifier/train.jsonl	teacher_training/data/processed/identifier/validation.jsonl	5,835	722
Local Structure	teacher_training/data/processed/local_structure/train.jsonl	teacher_training/data/processed/local_structure/validation.jsonl	11,085	1,385
Control Flow	teacher_training/data/processed/control_flow/train.jsonl	teacher_training/data/processed/control_flow/validation.jsonl	4,393	567

/home/tangyafeng/zhuxiyao/teacher-specialization formatting/test.jsonl identifier/test.jsonl local_structure/test.jsonl control_flow/test.jsonl

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Formatting Identifier Local Structure Exact Match Control Flow Pass@1/Test Pass

		Base	LoR	
Formatting	Exact Formatting Repair	38.26%	88.60%	+50.34 pp
Identifier	Exact Identifier Recovery	24.73%	51.79%	+27.06 pp
Local Structure	Exact Match	84.42%	99.78%	+15.37 pp
Control Flow	Pass@1 / Test Pass	63.48%	97.38%	+33.90 pp

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	Compile Base	Compile LoR	Parse Base	Parse LoR	Signature Base	Signature LoR
Formatting	99.46%	99.86%	99.46%	99.86%	99.32%	99.86%
Identifier	99.31%	100.00%	99.31%	100.00%	97.66%	100.00%
Local Structure	99.49%	100.00%	99.49%	100.00%	99.49%	100.00%
Control Flow	99.06%	98.88%	99.06%	98.88%	98.88%	98.88%

Formatting Levenshtein Distance 21.23 2.74 Identifier Subtoken F1 0.3681 0.6034 Local Structure CodeBLEU 0.9847 0.9998 Control Flow CodeBLEU 0.7301 0.7737

6 结果分析

5.1 Formatting

Formatting 38.26% 88.60% Compile Parse Signature Pass 99.86%

5.2 Identifier

Identifier 24.73% 51.79% Subtoken F1 0.3681 0.6034 Compile Parse Signature Pass 100%

5.3 Local Structure

Local Structure 84.42% 99.78% 15.58% 0.22% 98.6%

5.4 Control Flow

Control Flow Pass@1/Test Pass 63.48% 97.38% 33.90 LoRA Compile Parse Signature Pass 98.88% 0.19

7 失败案例与局限性

Control Flow 2048 token

paired bootstrap paired sign test

8 实验产物与复现记录

Base/LoRA summary.json per_sample.csv checkpoint experts_summary.csv experts_summary.json experiment_manifest.csv experiment_manifest.json

9 总结

Formatting Control Flow Identifier Subtoken F1 Local Structure LoRA